

SUPPLY CHAIN INTELLIGENCE TOOL

Global Rare Earth Supply Chain

Interactive Policy Scenario Modeler — Explainer

OVERVIEW

What This Tool Shows

This interactive visualization maps the complete rare earth element (REE) supply chain — from mining and extraction through chemical processing and separation, to end-use manufacturing in defense, clean energy, and consumer electronics. It is designed for government affairs teams, policy analysts, and supply chain strategists who need to understand where structural vulnerabilities exist and how policy decisions change the risk picture.

The tool is not a forecast. It describes concentration and flow patterns under different policy scenarios, and the same data point represents supply exposure for one actor and market position for another. Color and metric values indicate magnitude of change from baseline — not a judgment about preferred outcomes.

DATA & METHODOLOGY

Data Sources

The tool draws on the following primary sources:

- Production and reserve data from the USGS Mineral Commodity Summaries 2025
- Processing capacity estimates from CSIS Critical Minerals Security Program (2025)
- Demand projections from the IEA Critical Minerals Market Review 2024
- Company-level operational data from Lynas Rare Earths ASX filings and MP Materials SEC filings
- Western capacity buildout timelines from the Payne Institute for Public Policy (2025)
- Battery and magnet demand forecasts from BloombergNEF
- EU ReArm defense spending projections and US defense procurement data

Node sizes reflect relative production or processing capacity in thousand tons of rare earth oxide equivalent (kt REO). Flow line weights reflect relative trade volumes. All figures are estimates based on publicly available data; precise commercial volumes are not disclosed by most producers.

STRUCTURAL FACTS

Key Concentration Metrics — Baseline 2025

Metric	Baseline 2025
China rare earth mining share	~70%
China rare earth processing share	~90%
Non-China HREE separation capacity	~1,500 t/y
New producer material routed via China	>95%
2030 non-China capacity (BNEF est.)	~50% of CN
IRA FEOC exclusion threshold (EV)	2027

The core structural finding is that mining diversification without processing diversification is strategically irrelevant. Australia produces 13kt/year at Mt Weld and Myanmar supplies significant heavy rare earth (HREE) volumes — but the overwhelming majority of this material routes to Chinese processors before reaching end-use manufacturers. Non-Chinese HREE separation capacity outside China remains critically insufficient even under optimistic Western buildout scenarios.

SCENARIO MODELER

The Five Policy Scenarios

Baseline 2025	Current environment. Chinese export controls on processing chemicals in effect but not escalated. Lynas Malaysia operational; MP Materials separating LREEs.
Export Controls — Partial	Beijing extends controls to separated oxides and alloys. Flow disruption concentrated in finished material trade.
Export Controls — Full	Controls extended to raw ore, oxides, alloys, and magnets. Near-complete restriction of Chinese rare earth exports.
Western Capacity Expansion	IRA FEOC exclusions enforce on schedule. Lynas Texas and Iluka Eneabba commission. Policy-intent scenario — gap from baseline is execution risk.
Chinese Processing Integration	New producers from Africa and Southeast Asia route through Chinese processors under long-term offtake agreements.

Each scenario updates the map's flow lines, node visibility, and four concentration metrics simultaneously. The metrics measure Chinese supply share for US Defense, EU Defense, Western OEM FEOC regulatory exposure, and the share of new producer output routed through Chinese processing. A longer bar indicates higher concentration.

ANALYTICAL CONCLUSIONS

What the Data Shows

The visualization supports **four conclusions** that run against the prevailing policy narrative:

1. Processing is the chokepoint, not mining.

Western diversification efforts focus heavily on mining — Mountain Pass, Mt Weld, potential African sources. But without corresponding processing capacity, diversified mining simply feeds the same Chinese refineries. The chokepoint is the separation and refining stage, not the ground.

2. The chokepoint degrades once deployed.

Chinese export controls, once deployed, function as the starting gun for the countdown to their own obsolescence. They trigger substitution investment — in Lynas Texas, Iluka Eneabba, Energy Fuels — that would not otherwise occur at speed. The Export Controls — Full scenario shows maximum leverage; it also shows maximum incentive to build around it.

3. HREE separation remains the critical gap.

Light rare earth (LREE) separation outside China is now commercially operational — Lynas Malaysia, MP Materials California. Heavy rare earth (HREE) separation is a different story. The ~1,500 t/y of non-Chinese HREE separation capacity is insufficient for defense applications alone, let alone clean energy scale-up. This is where execution risk is highest in the Western Capacity Expansion scenario.

4. The 2026+ demand trajectory changes the calculus.

The EU ReArm program (3.5x rare earth demand multiplier for defense applications) and sustained US defense procurement are shifting the demand mix. The 2026+ Trajectory toggle shows how a defense-weighted demand structure increases dependency metrics relative to the 2023 commercial baseline — particularly for HREE-intensive applications like permanent magnets in precision guidance systems.

Access the tool at marketaccess-x.com/supply-chain-mapping

For bespoke supply chain analysis, scenario modelling for specific sectors, or briefings contact us at marketaccess-x.com.